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Bewijs dat $\text{Bgtan } \frac{1}{2} = \text{Bgtan } \frac{1}{3} + \text{Bgtan } \frac{1}{7}$

$$\text{REchterzijde} = \text{Bgtan} \left(\frac{\frac{1}{3} + \frac{1}{7}}{1 - \frac{1}{3} \cdot \frac{1}{7}} \right)$$

$$= \text{Bgtan} \left(\frac{\frac{10}{21}}{1 - \frac{1}{21}} \right)$$

$$= \text{Bgtan} \left(\frac{10}{21} \cdot \frac{20}{21} \right)$$

$$= \text{Bgtan} \left(\frac{1}{2} \right)$$

$$\text{Linkerzijde} = \text{Bgtan} \left(\frac{1}{2} \right)$$

$$\Rightarrow \text{L.L} = \text{R.L}$$

Q.E.D.

References